计算方法预测三级结构：穿线法 I-TASSER

虽然同源建模是蛋白质结构预测的首选方法，但是对于那些找不到合适模板（一致度大于 30%的模板）的蛋白质，此方法并不适用。这种情况下可以尝试穿线法（threading）。穿线法基于的原理是：不相似的氨基酸序列也可以对应着相似的蛋白质结构。

PDB 数据库里已解析出的真实蛋白质结构每天都会增加，但是从 2008 年开始就再没有新的结构拓扑产生了（图 1）。换言之，目前发现的蛋白质结构的种类已定格在 2008 年的 1393 种了，即已知的 10 万多个结构，根据结构拓扑划分，分成了 1393 种。新解析出来的结构必定属于其中之一。而具有同一结构拓扑的蛋白质，序列水平上有相似的，也有不相似的。

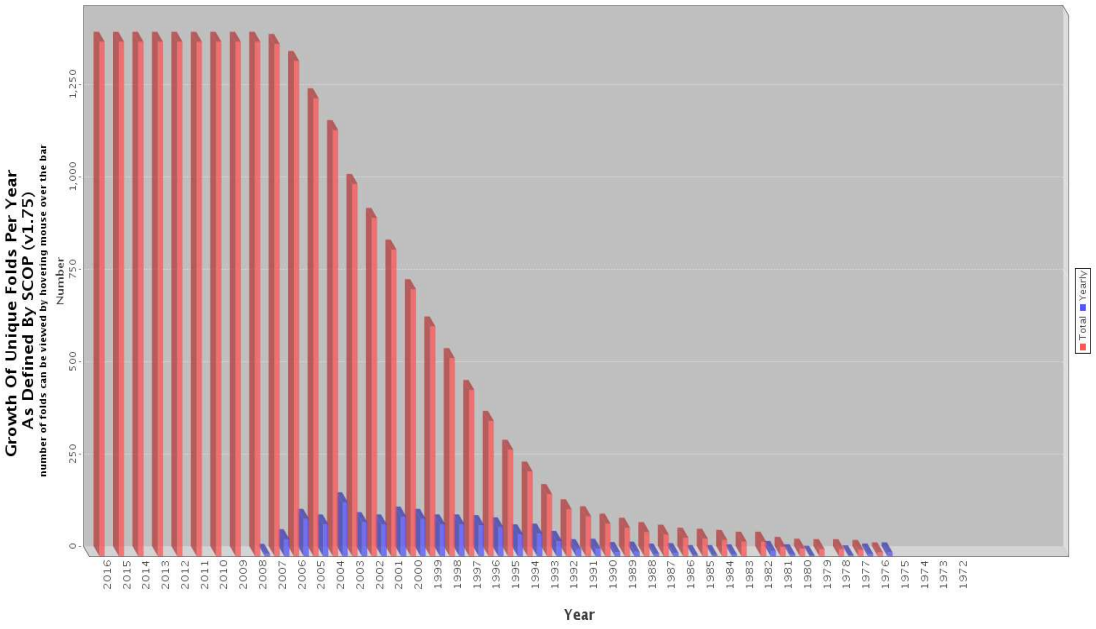


图 1.PDB 数据中解析结构的结构拓扑数量年增长情况

图片来源：http://www.rcsb.org/pdb/statistics/contentGrowthChart.do?content=fold-scop

在上述前提下，我们可以把目标序列像线一样穿到目前现有的结构里，看穿到哪个结构里最舒服，哪个结构就可以作为预测的模板，并根据最舒服的穿法，构建出最终模型。那怎么知道穿的舒服不舒服呢？通过能量方程。穿的舒服，能量就低，穿的不舒服，能量就高。这和我们穿衣服一样。穿上一件不合身的衣服，你肯定老在那扭啊扭的不得劲，这老动换能量就高啊！要是穿上件合身的，那就能待住不动了，这能量不就降下来了。穿线法就是通过计算目标序列穿到每一个已知结构中的每一种穿法下的能量，找到能量最低的那种穿法以及所穿的结构，然后把目标序列中的氨基酸替换到模板结构里来构建结构模型的（图 2）。显

然这种方法的计算量较同源建模法要大得多，因此预测需要耗费更久的时间。

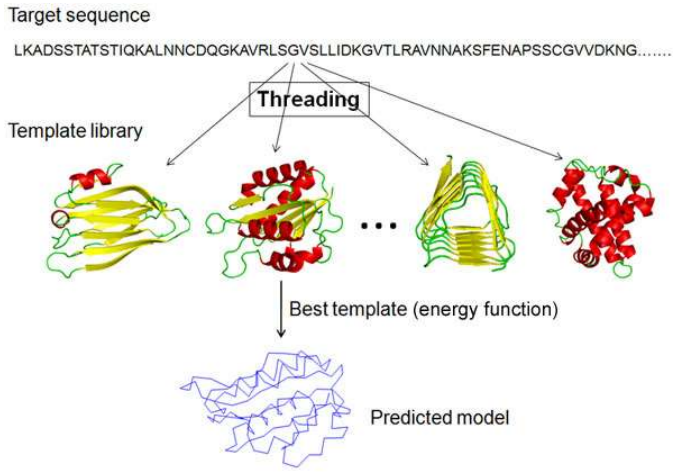


图 2. 穿线法基本原理

用穿线法预测蛋白质三级结构，首选密歇根大学张阳教授的 I-TASSER（http://zhanglab.ccmb.med.umich.edu/I-TASSER）。这个预测系统在 CASP 第 7 到第 11 届蛋白质结构预测比赛中都名列第一。CASP 全称是蛋白质结构生物信息预测国际竞赛。两年一届。每次比赛，参与者们会对一组即将公开的结构进行预测。再将预测模型和真实结构进行比较，看谁预测的最准。咱们中国上海交大电子信息与电器工程学院的沈红斌教授的研究组在 CASP11 中取得了第三名。这是中国代表队截至目前取得的最好成绩。

张阳教授的 I-TASSER 可以在线提交预测任务（图 3），不需要提前下载安装。提交氨基酸序列进行预测前需注册获得用户名密码，注册是完全免费的。再给任务起个名字。最后点 Run I-TASSER。

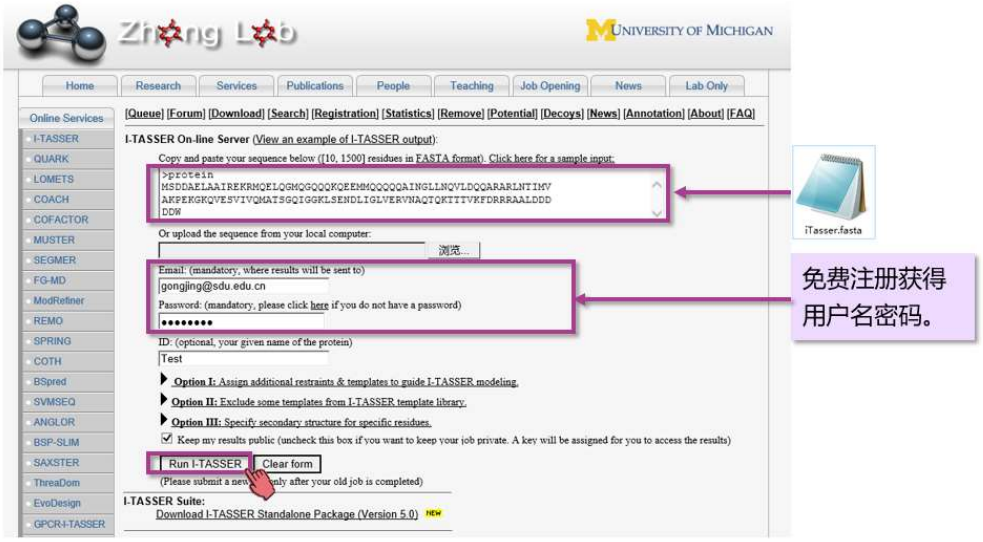


图 3. I-TASSER 提交页面

任务提交之后，请记住任务号或保存结果网址，以便日后查看。此外，通过点击 Queue 链接，可以查看当前的所有任务进程（图 4）。从图 5 的任务列表里可以看出，穿线法需要的时间确实比同源建模法多得多，提交的这条示例序列需要计算大约 35 个小时。这里请注意，一个用户或一个 IP 地址一次只能提交一个任务。



图 4. 查找已提交任务

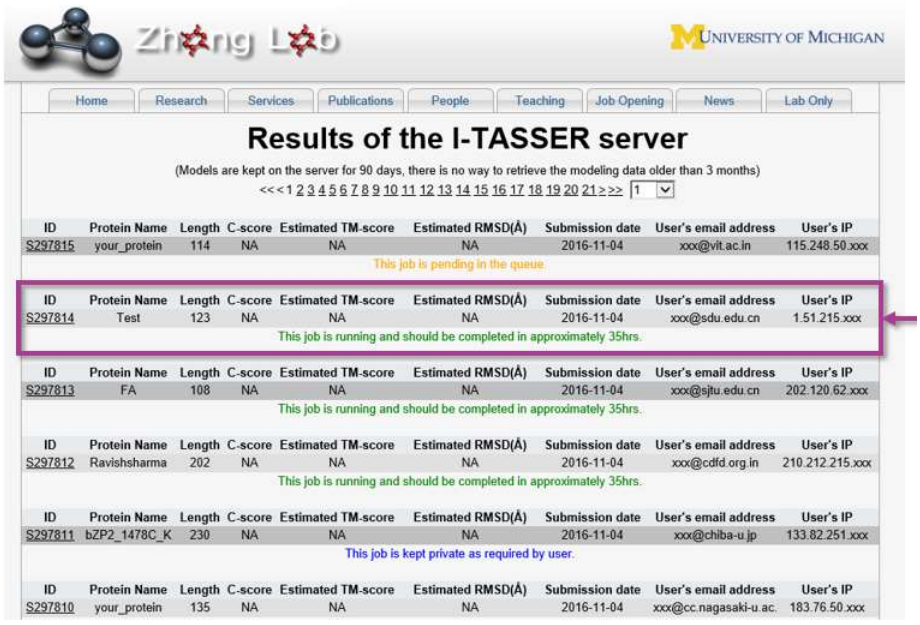


图 5. Queue 页面查看所有任务

此外，还可以通过搜索链接，搜索任务号找到任务，或者搜索账号找到账号下所有的任务，或者通过搜索序列查找（图 6）。

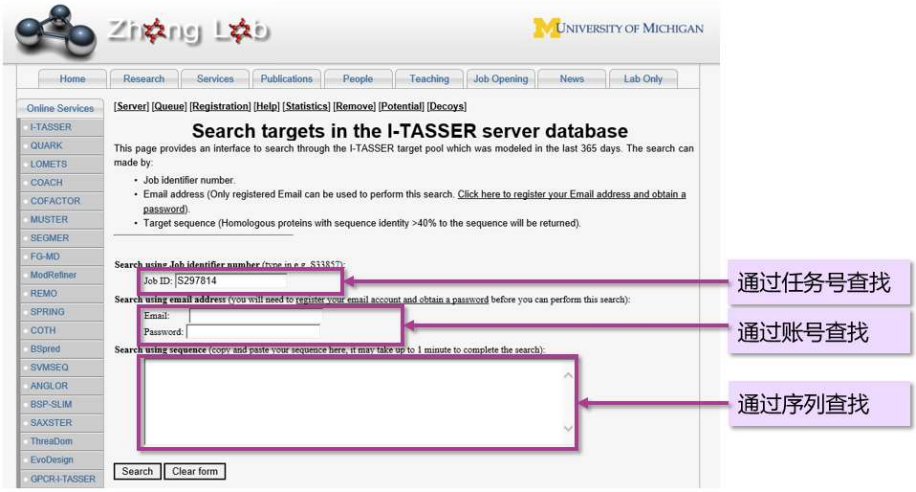


图 6. Search 页面查找已提交任务

近 40 个小时之后，结果终于出来了（图 7）。ITASSER 预测出了 5 个模型，点击任务号以查看具体信息。

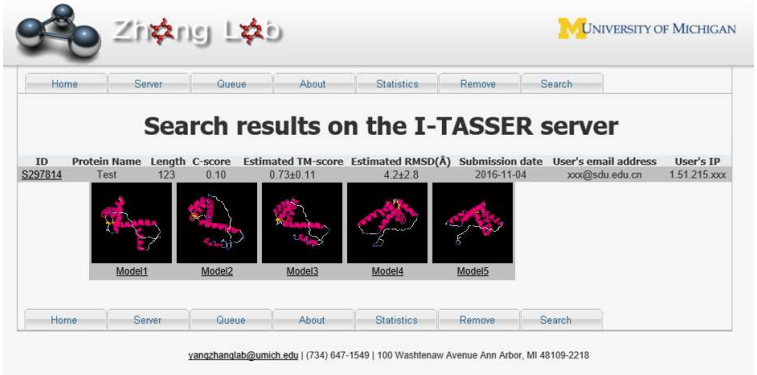


图 7. 任务完成列表页面

I-TASSER 除了预测出了三级结构，同时还预测除了二级结构和残基可溶性（图 8）。

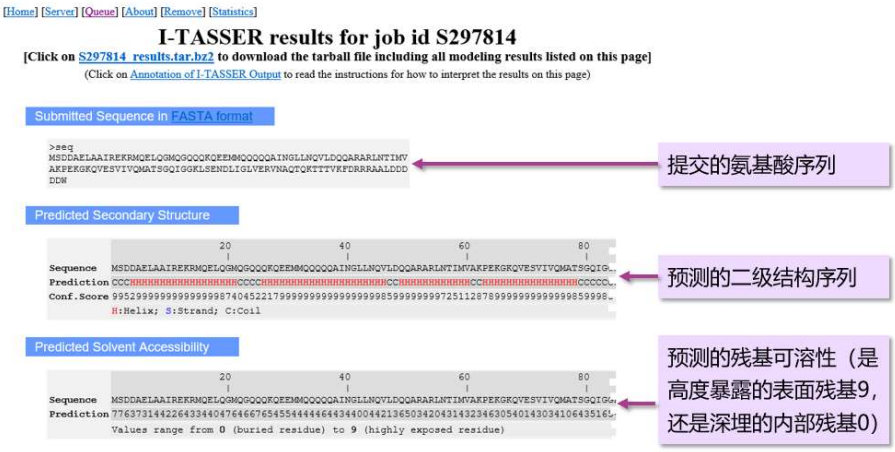
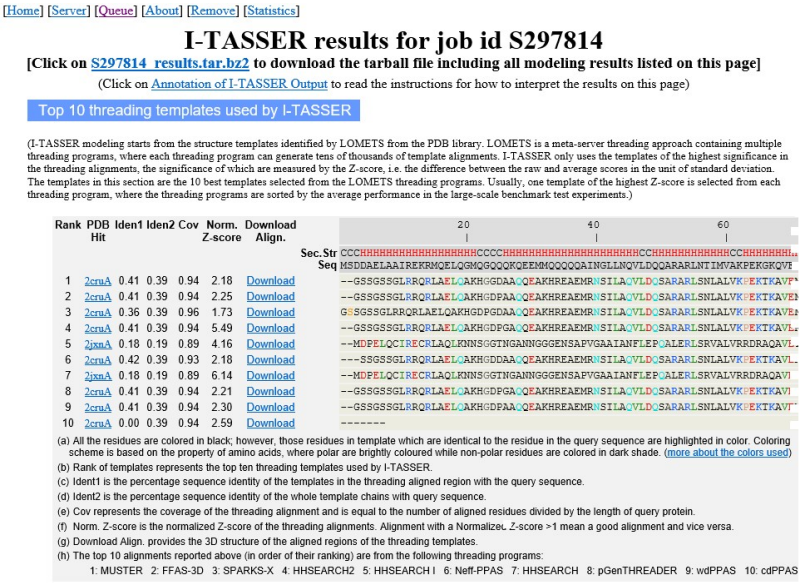


图 8. 结果页面 1

图 9 为构建模型所使用的模板及序列比对。这些模板和序列别对不是通过序列相似性创建的，而是用穿线法穿出来的。


[Home] [Server] [Queue] [About] [Remove] [Statistics]

I-TASSER results for job id S297814

[Click on S297814_results.tar.bz2 to download the tarball file including all modeling results listed on this page]

(Click on Annotation of I-TASSER Output to read the instructions for how to interpret the results on this page)

Top 10 threading templates used by I-TASSER

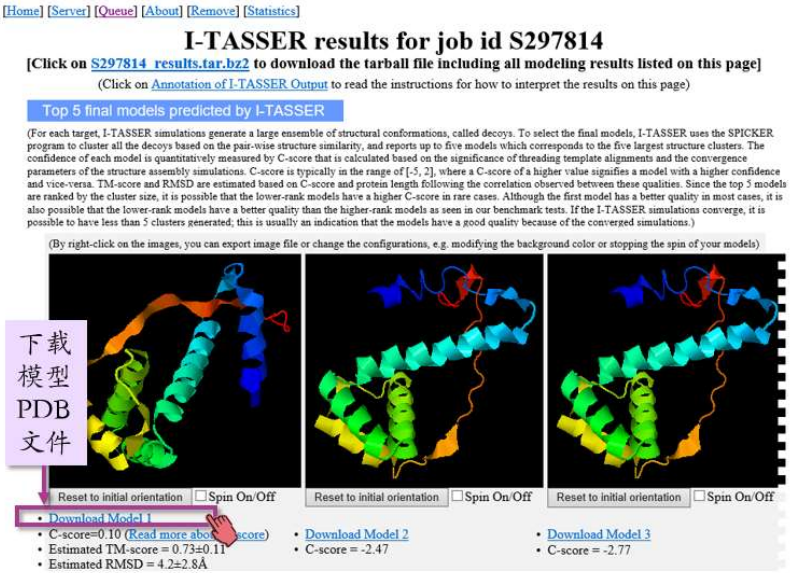
(I-TASSER modeling starts from the structure templates identified by LOMETS from the PDB library. LOMETS is a meta-server threading approach containing multiple threading programs, where each threading program can generate tens of thousands of template alignments. I-TASSER only uses the templates of the highest significance in the threading alignments, the significance of which are measured by the Z-score, i.e. the difference between the raw and average scores in the unit of standard deviation. The templates in this section are the 10 best templates selected from the LOMETS threading programs. Usually, one template of the highest Z-score is selected from each threading program, where the threading programs are sorted by the average performance in the large-scale benchmark test experiments.)

Rank	PDB Hit	Iden1	Iden2	Cov	Norm. Z-score	Download Align.
1	2cruA	0.41	0.39	0.94	2.18	Download
2	2cruA	0.41	0.39	0.94	2.25	Download
3	2cruA	0.36	0.39	0.96	1.73	Download
4	2cruA	0.41	0.39	0.94	5.49	Download
5	2jxnA	0.18	0.19	0.89	4.16	Download
6	2cruA	0.42	0.39	0.93	2.18	Download
7	2jxnA	0.18	0.19	0.89	6.14	Download
8	2cruA	0.41	0.39	0.94	2.21	Download
9	2cruA	0.41	0.39	0.94	2.30	Download
10	2cruA	0.00	0.39	0.94	2.59	Download

(a) All the residues are colored in black; however, those residues in template which are identical to the residue in the query sequence are highlighted in color. Coloring scheme is based on the property of amino acids, where polar are brightly coloured while non-polar residues are colored in dark shade. (more about the colors used)
(b) Rank of templates represents the top ten threading templates used by I-TASSER.
(c) Iden1 is the percentage sequence identity of the templates in the threading aligned region with the query sequence.
(d) Iden2 is the percentage sequence identity of the whole template chains with query sequence.
(e) Cov represents the coverage of the threading alignment and is equal to the number of aligned residues divided by the length of query protein.
(f) Norm. Z-score is the normalized Z-score of the threading alignments. Alignment with a Normalized Z-score >1 mean a good alignment and vice versa.
(g) Download Align. provides the 3D structure of the aligned regions of the threading templates.
(h) The top 10 alignments reported above (in order of their ranking) are from the following threading programs:
1: MUSTER 2: FFAS-3D 3: SPARKS-X 4: HHSEARCH2 5: HHSEARCH I 6: Neff-PPAS 7: HHSEARCH 8: pGenTHREADER 9: wdPPAS 10: cdPPAS

图 9. 结果页面 2

像许多预测软件一样，ITASSER 给出了多个预测模型（图 10）。这些模型根据 ITASSER 自带的质量评估系数 C-score 排序。C-score 从负 5 到正 2，分值越高模型越可信。除了 C-Score，排名第一的模型还给出了一个 TM-Score，这也是评价模型质量的系数。TM-socre >0.5 说明模型具有正确的结构拓扑，可信；<0.17 说明模型属于随机模型，不可信。挑选到合格的模型后，点击 download model 下载模型的 PDB 文件。


[Home] [Server] [Queue] [About] [Remove] [Statistics]

I-TASSER results for job id S297814

[Click on S297814_results.tar.bz2 to download the tarball file including all modeling results listed on this page]

(Click on Annotation of I-TASSER Output to read the instructions for how to interpret the results on this page)

Top 5 final models predicted by I-TASSER

(For each target, I-TASSER simulations generate a large ensemble of structural conformations, called decoys. To select the final models, I-TASSER uses the SPICKER program to cluster all the decoys based on the pair-wise structure similarity, and reports up to five models which corresponds to the five largest structure clusters. The confidence of each model is quantitatively measured by C-score that is calculated based on the significance of threading template alignments and the convergence parameters of the structure assembly simulations. C-score is typically in the range of [-5, 2], where a C-score of a higher value signifies a model with a higher confidence and vice-versa. TM-score and RMSD are estimated based on C-score and protein length following the correlation observed between these qualities. Since the top 5 models are ranked by the cluster size, it is possible that the lower-rank models have a higher C-score in rare cases. Although the first model has a better quality in most cases, it is also possible that the lower-rank models have a better quality than the higher-rank models as seen in our benchmark tests. If the I-TASSER simulations converge, it is possible to have less than 5 clusters generated; this is usually an indication that the models have a good quality because of the converged simulations.)

(By right-click on the images, you can export image file or change the configurations, e.g. modifying the background color or stopping the spin of your models)

下载模型 PDB 文件

Reset to initial orientation | Spin On/Off | Reset to initial orientation | Spin On/Off | Reset to initial orientation | Spin On/Off

- Download Model 1
- C-score=0.10 (Read more about C-score)
- Estimated TM-score = 0.73±0.11
- Estimated RMSD = 4.2±2.8Å

- Download Model 2
- C-score = -2.47

- Download Model 3
- C-score = -2.77

图 10. 结果页面 3

此外，I-TASSER 还为预测了该蛋白质的功能（图 11）。它把排名第一的模型和 PDB 中的结构进行了比较，把跟模型最相似的 PDB 结构找了出来。这样就可以通过这些已知结构的功能他推测模型的功能了。

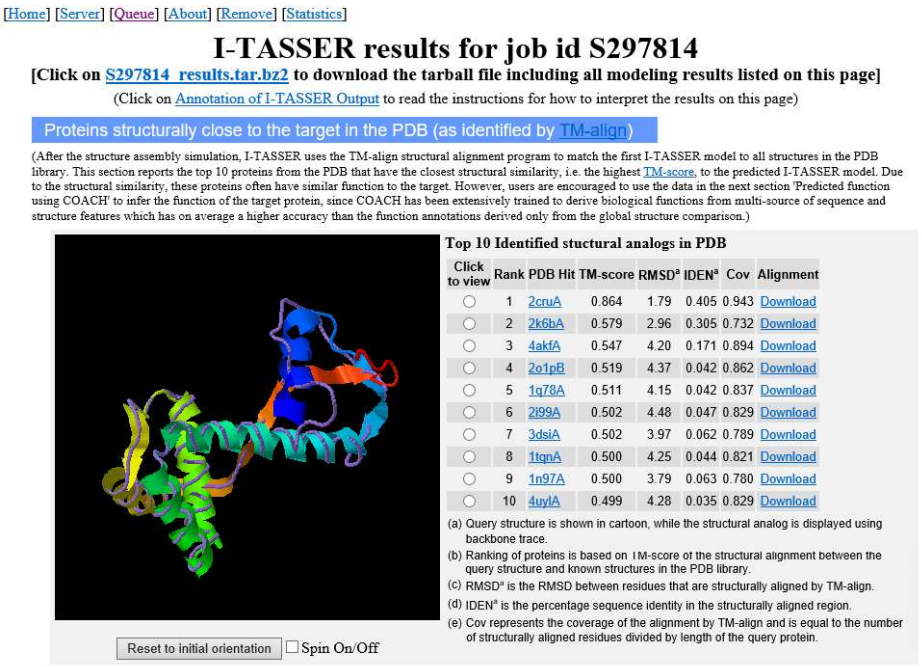


图 11. 结果页面 4

I-TASSER 还预测出了模型可能结合的配体以及具体的配体结合位点。并给出了可能的功能（图 12）。

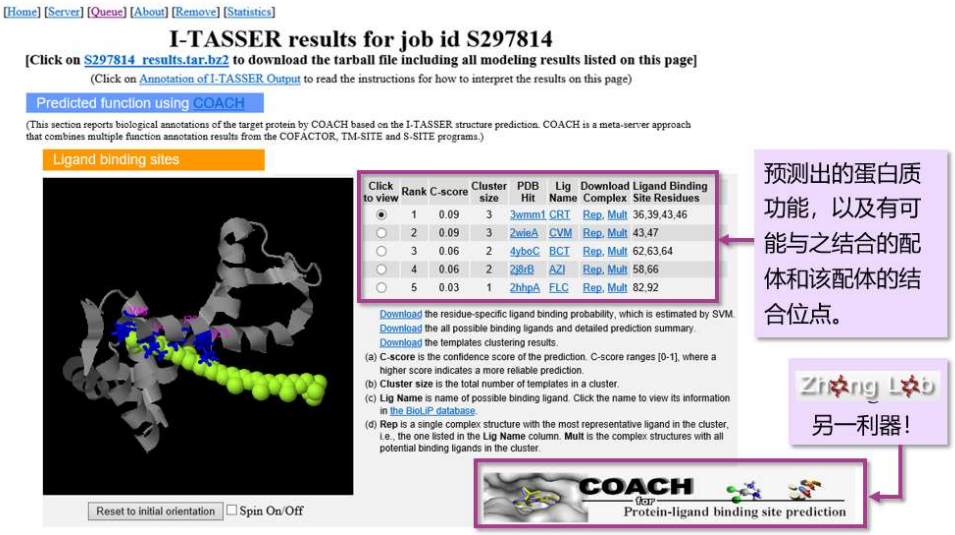


图 12. 结果页面 5